

RESEARCH ARTICLE

Dual-Mode CMOS LIF Neuron With Subthreshold Efficiency and Saturation-Driven Robustness

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ABSTRACT We present a compact CMOS leaky integrate-and-fire neuron that operates natively in two regimes: subthreshold for ultra-low power and saturation for robust interfacing. This dual-mode functionality bridges the energy efficiency of the subthreshold operation with the connectivity advantage of saturation within a single circuit. The design leverages device-level parasitics and subthreshold conduction to minimize the energy consumption per spike and reduce silicon area. It also maintains sufficient headroom in the saturation region to drive or directly interface with non-subthreshold components and resistive memories. In addition to outlining the design methodology and sizing strategy that enable dual-mode operation without needing additional amplification stages, we introduce a mathematical model that captures the circuit dynamics. Under specified operating assumptions, this model allows us to predict the spiking frequency for various application targets. SPICE simulations validate the expected spiking dynamics and trade-offs between energy consumption and latency. We also compare our proposed neuron with recent CMOS implementations reported in the literature. While some earlier designs achieve lower absolute energy consumption, our approach features a significantly reduced transistor count and the ability to interface with other cells without dedicated amplification, offering a practical and integration-friendly solution. By converting unavoidable capacitive parasitics into advantages, this work presents a silicon-efficient method for low-power neuromorphic hardware, making both subthreshold efficiency and saturation-range connectivity key design objectives.

INDEX TERMS Analog VLSI, CMOS, leaky integrate-and-fire (LIF), low-power neuromorphics, neuromorphic hardware, saturation connectivity, subthreshold efficiency.

I. INTRODUCTION

The growing demand for portable and autonomous devices has intensified the need to optimize the energy consumption in electronic systems. In this context, while artificial neural networks have shown remarkable versatility, running them in software on general-purpose processors or programmable accelerators entails a high computational load and elevated power consumption [1], [2], which limits their adoption in

embedded scenarios or under tight energy budgets. This landscape has motivated specialized hardware implementations, including analog/digital neuromorphic hardware, aimed at reducing energy consumption and silicon area while preserving learning or inference functionality. Moreover, recent studies indicate that the energy efficiency of artificial neurons is strongly influenced by parameters such as membrane capacitance and operating voltage, and that very low power levels can be achieved when transistors operate in a deep-subthreshold regime [3]. Nevertheless, subthreshold circuits exhibit inherent limitations due to

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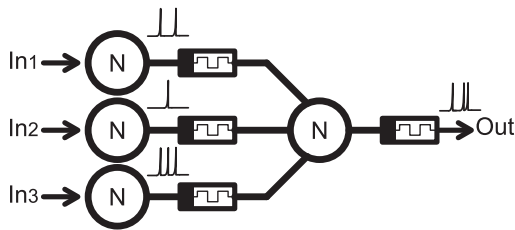


FIGURE 1. Conceptual illustration of a spiking neural network implemented with memristive synapses and CMOS LIF neurons. The neuron output operates in the saturation range.

their low voltage supply levels [4], making them highly susceptible to noise and constraining their analog operating range. Furthermore, direct interconnection with other blocks operating in saturation can pose challenges (such as analog processing stages or digital interfaces) and often necessitates additional coupling or amplification stages to ensure voltage-level compatibility.

Circuits designed to operate exclusively in the saturation region offer several advantages, including enhanced linearity, improved noise robustness, and a wider usable voltage swing. These characteristics make them particularly well suited for seamless integration with both analog and digital components. However, such benefits often come at the expense of increased circuit complexity, as they typically require a higher transistor count, which in turn leads to greater power consumption [5], [6], [7]. This inherent trade-off between energy efficiency in subthreshold operation and I/O robustness in saturation has become a central theme in the design of neuron circuits for neuromorphic hardware.

This work presents a compact analog architecture for a leaky integrate-and-fire (LIF) spiking neuron model. The proposed model uses a mixed subthreshold and saturation operation, combining the benefits of both approaches. It takes advantage of the low energy consumption per spike event, which is characteristic of subthreshold operation while maintaining the rail-to-rail output swing and, therefore, the level compatibility offered by saturation. The main concept is to utilize a single subthreshold-operated device for charge and leak integration while maintaining the other stages in saturation for functions such as thresholding, spiking, resetting, and input/output operations. This approach enables a functional implementation with fewer transistors, eliminating the need for additional coupling stages or multiple supply rails. Avoiding these stages reduces typical area and power overheads, facilitating plug-and-play integration with saturation-range analog and digital blocks, as well as with platforms based on resistive memories [8] (Fig. 1).

A. ABBREVIATIONS AND ACRONYMS

The following abbreviations are used in this manuscript: CMOS — complementary metal–oxide–semiconductor;

DRC — design–rule–checking; LIF — leaky integrate-and-fire; MOSFET — metal–oxide–semiconductor field-effect transistor; PDK — process design kit; SPICE — simulation program with integrated circuit emphasis; VTC — voltage transfer characteristic.

B. PAPER ORGANIZATION

The remainder of this paper is organized as follows: **Section II** introduces the mathematical model of the LIF neuron, which serves as the analytical baseline for our circuit design. It presents the differential equation governing membrane charging and reset, defines the key state variables and parameters, and establishes the operating assumptions. This section also summarizes MOS transistor behavior in the subthreshold region.

Section III presents the proposed circuit in detail and explains its operation across three functional stages: (i) integration/reset, (ii) threshold and spike generation, and (iii) gain- and feedback-based reset. We derive the sizing-relevant expressions, specifically, transistor W/L ratios used to parameterize the design, including the inverter trip point that defines the LIF threshold, the effective leakage resistance in the subthreshold region, the membrane capacitance (accounting for intrinsic and parasitic contributions as well as any explicit capacitors), and the available excitation current options. Building on these relations, we establish a closed-form relationship between the trip point, the average leakage resistance, and the membrane capacitance to estimate the target firing frequency.

Section IV reports the results and discussion. We compare theoretical predictions with SPICE simulations (both pre- and post-layout) in the SKY130 process, quantify the level of agreement in firing frequency across operating points, and analyze the energy–latency trade-offs enabled by the proposed dual-mode topology. This section also reports the silicon area obtained from the layout, discusses the impact of parasitics on firing frequency, particularly the post-layout SPICE simulation slowdown, and presents a comparative table against recent LIF implementations reported in the literature (both subthreshold-only and saturation-only), with emphasis on transistor count, energy per spike, and interface requirements.

Section V concludes the paper by summarizing the main findings and limitations and outlining future improvements toward application-specific tuning.

II. THEORY AND MODELING

In this section, the mathematical model of the LIF is presented as the analytical basis for the circuit design, complemented by an overview of transistor behavior in the subthreshold region.

A. LIF MODEL

The LIF neuron, as illustrated in Fig. 2, consists of a membrane capacitor C_m , a leak resistor R_{leak} , and an excitatory input current signal I_{ex} . Under a constant I_{ex} , the membrane voltage, V_m , evolves according to a first-order

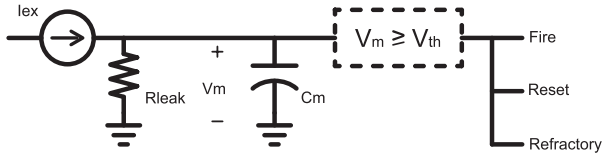


FIGURE 2. Equivalent electrical model of the leaky integrate-and-fire (LIF) neuron, showing the membrane capacitance C_m , the leakage resistance R_{leak} , the excitation current I_{ex} , and the firing/reset mechanism.

linear dynamics with a time constant $\tau = R_{leak}C_m$. When V_m reaches a threshold voltage V_{th} , the neuron generates a spike event at the output node, denoted by V_{out} . Following the spike event, the membrane voltage is reset to V_{reset} , and the integration process resumes.

The charge–discharge dynamics of the membrane node are described by (2), (1).

$$C_m \frac{dV_m}{dt} + \frac{V_m}{R_{leak}} - I_{ex} = 0, \quad (1)$$

where,

$$I_{ex} = I_{Cm} + I_R. \quad (2)$$

I_{Cm} , and I_R , are the currents flowing through R_{leak} and C_m . The equation in (1) has been widely used in analog hardware implementations of LIF neurons [9], for which a general solution is given by (3):

$$v_m(t) = R_{leak}I_{ex} + \frac{k}{\exp(t/(R_{leak}C_m))}, \quad (3)$$

where k is set by the initial condition $V_m(0) = V_{reset}$, yielding $k = V_{reset} - R_{leak}I_{ex}$. Therefore, the membrane voltage evolves as in (4):

$$v_m(t) = R_{leak}I_{ex} + (V_{reset} - R_{leak}I_{ex}) e^{-t/(R_{leak}C_m)}. \quad (4)$$

When $v_m(t)$ reaches V_{th} , the neuron fires, and the membrane potential is reset to V_{reset} , closing the integrate–fire–reset cycle. For compactness, we reference the LIF dynamics with the piecewise expression in (5), where $\tau = R_{leak}C_m$:

$$v_m(t) = \begin{cases} R_{leak}I_{ex}(1 - e^{-t/\tau}), & V_m < V_{th(lif)}, \\ \text{fire, reset, and refractory,} & V_m \geq V_{th(lif)}. \end{cases} \quad (5)$$

B. SUBTHRESHOLD CONDUCTION

In digital electronics, MOS transistors are commonly operated in strong inversion to toggle between the logical states of cut-off and saturation. However, describing a device as ‘off’ is a simplifying abstraction that doesn’t reflect the underlying physics of the device. From a physical standpoint, even when the gate-to-source voltage is below threshold, the transistor conducts a small but finite current [10], [11], [12], governed by carrier diffusion along the channel rather than by the drift-dominated transport characteristic of strong inversion. While drift-current mechanisms in MOS transistors are extensively treated in the literature [13],

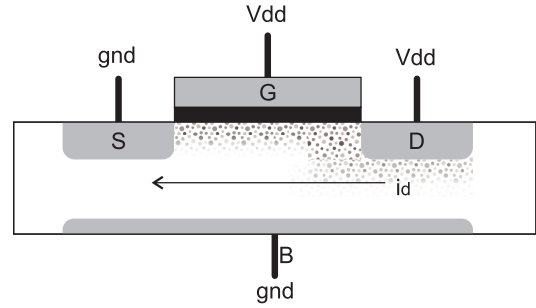


FIGURE 3. Illustration of drift-dominated carrier transport in strong inversion, where charge motion is primarily driven by the electric field between drain and source., in the strong inversion operation.

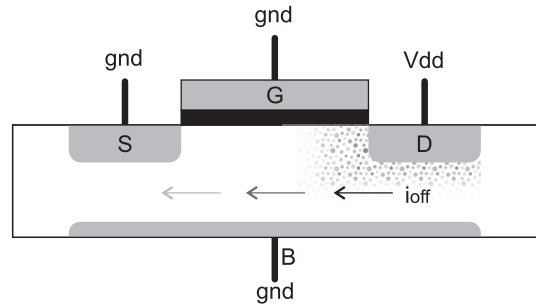


FIGURE 4. Illustration of diffusion-dominated carrier transport in subthreshold operation, where current arises from carrier concentration gradients even for $V_{GS} < V_{th}$.

[14], the diffusion-driven conduction that dominates in the subthreshold regime is often given comparatively less emphasis in circuit-level expositions.

When a MOS transistor operates under strong electric fields, such as in the case of strong inversion, both electrons (the negative carriers in N-MOS transistors) and holes (the positive carriers in P-MOS transistors) are primarily influenced by the applied field between the drain and source terminals. Electrons drift toward the positive drain terminal, while holes drift toward the negative source terminal. This movement, driven by the electric field, is referred to as *drift current* (Fig. 3). By contrast, *diffusion current* arises from carrier concentration gradients. Regions with higher electron/hole concentrations tend to spread into regions of lower concentration, producing a net current even in the absence of a strong external electric field (Fig. 4).

This behavior is modeled by the subthreshold drain–current expression as in (7) [15], where the current does not vanish for $V_{gs} < V_{th}$; instead, it decreases exponentially as V_{gs} is reduced. Consequently, even for $V_{gs} = 0$ [11], a small (yet nonzero) subthreshold leakage current is present, as in (7), confirming that a MOSFET is never fully “off.”

$$I_0 = \mu_0 C_{ox} \frac{W}{L} (n - 1) V_t^2, \quad (6)$$

$$I_d = I_0 \exp\left(\frac{V_{gs} - V_{th}}{n V_t}\right) \left(1 - e^{-V_{ds}/V_t}\right). \quad (7)$$

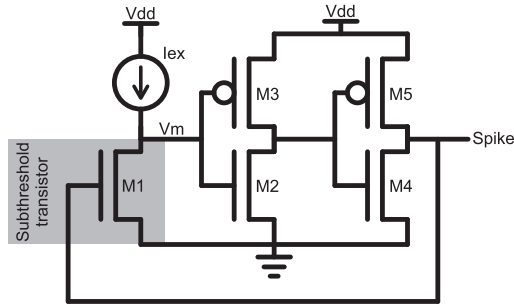


FIGURE 5. Proposed dual-mode CMOS LIF neuron topology. Transistor M1 operates in the subthreshold region to perform leaky integration, while transistors M2–M5 operate in saturation to provide threshold detection, spike generation, gain, and reset with full-swing output.

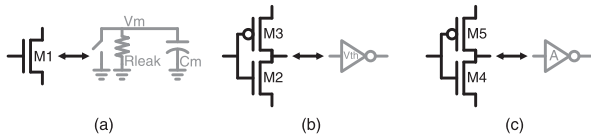


FIGURE 6. Functional operating stages of the proposed LIF neuron: (a) integration and reset model at the membrane node, (b) threshold detection and spike generation, and (c) gain and feedback-based reset stage.

In summary, when a MOS transistor operates in the subthreshold region, it is not truly turned off; instead, it conducts a small current dominated by carrier diffusion, which, from a circuit-level perspective, manifests as a very large effective resistance at the membrane node, V_m .

III. CIRCUIT DESIGN AND OPERATION

This section describes the proposed circuit and its operation across the functional stages, highlighting the key expressions and parameters that enable estimation of the firing frequency.

A. CIRCUIT OPERATION

The proposed circuit (Fig. 5) is organized into three main stages: integration and reset (transistor M1, operated in subthreshold), threshold detection and spike generation (transistors M2–M3), and gain with feedback-based reset (transistors M4–M5). Next, we will explain each operating stage in detail.

- 1) **Integration and reset (Fig. 6(a)):** The excitation current I_{ex} charges the membrane capacitance at the drain node of transistor M1, which is formed by the aggregate of intrinsic and parasitic capacitances connected to that node. As the membrane voltage V_m increases and reaches the inverter trip point $V_{th(lif)}$, which defines the membrane threshold and is set by transistors M2–M3, the subsequent stage of operation is activated.
- 2) **Threshold and spike (Fig. 6(b)):** The M2–M3 inverter senses V_m and toggles abruptly at its switching threshold $V_{th(lif)}$, producing the spike that drives the following stage.

- 3) **Gain and feedback reset (Fig. 6(c)):** The M4–M5 inverter stage provides the final output and feeds back a short reset pulse. Upon receiving this feedback, M1 is momentarily driven into conduction, enabling the leakage path and discharging C_m . Once the pulse ends, M1 returns to its subthreshold integration role, starting a new charge cycle and thus yielding a periodic spike train. In addition, the neuron's output voltage, V_{out} , reaches digital-compatible levels, up to 1.8 V for SKY130 nominal transistors (Fig. 7(c)).

This feedback mechanism yields stable oscillatory behavior without the need for external passive components, relying solely on the device's intrinsic capacitances.

B. LIF CIRCUIT MODEL

The proposed circuit consists of five transistors and an excitation current source, as shown in Fig. 5. Transistor M1 performs the integration of the input excitation current I_{ex} , feed at the neuron input. It exploits the parasitic capacitances present at its drain node, together with its intrinsic channel resistance in the subthreshold region, to establish an effective membrane time constant $\tau = R_{leak}C_m$, as long as the membrane voltage V_m remains below the inverter trip point $V_{th(lif)}$. Once the gate voltage exceeds this threshold, M1 transitions to a strong-conduction regime and rapidly discharges the accumulated charge at the membrane node, thereby implementing the reset operation (see Fig. 6(a)). To quantify the leakage resistance associated with the V_m node in the subthreshold regime, denoted as R_{leak} , it is necessary to recall the subthreshold drain-current model given in (7) [16]. Operating M1 in subthreshold is key because, between spike events, the device behaves as a controlled-leaky element, providing a well-defined, tunable effective resistance that governs the membrane dynamics prior to threshold crossing.

Additionally, the gate signal switches rapidly between voltage rails. This transition must happen quickly; as soon as the membrane voltage exceeds $V_{th(lif)}$, the M1 channel immediately drains the charge stored at its drain node. Since the integration currents are in the range of nanoamperes, R_{leak} needs to be quite large (typically in the $M\Omega$ to $G\Omega$ range) to achieve a practical charging time, which is generally in the tens to hundreds of microseconds.

$$R_{leak} = \left| \frac{V_{ds}}{i_d} \right| = \left| \frac{V_{ds}}{I_0 \exp\left(\frac{V_{gs} - V_{th}}{n V_t}\right) (1 - e^{-V_{ds}/V_t})} \right|. \quad (8)$$

The last expression (8) is significant because it indicates that the resistance is influenced by both the drain-source voltage and the gate-source voltage. However, during the intervals between spikes, the gate-source voltage is considerably smaller than the N-MOS transistor threshold V_{th} , allowing us to approximate $V_{gs} \approx 0$. Furthermore, since

$V_{ds} = V_m$, the equation (8) simplifies to:

$$R_{leak} = \left| \frac{V_m}{I_0 \exp\left(-\frac{V_{th}}{nV_t}\right) (1 - e^{-V_m/V_t})} \right|. \quad (9)$$

As noted, the leak resistance R_{leak} depends primarily on the membrane potential V_m . Since V_m varies over time during the integration phase, R_{leak} is not constant. By inspecting the capacitor-voltage plot (Fig. 7(a)) and the R_{leak} plot (Fig. 7(b)), two relevant regions can be distinguished:

- **Positive voltage** ($0 \leq V_m \leq V_{th(lif)}$): R_{leak} increases almost monotonically (approximately linearly over the range of interest), from a minimum value, when $V_m \approx 0$, to a maximum value, as $V_m \approx V_{th(lif)}$. Once the threshold is exceeded, the inverter switches, changes the device's operating regime, and discharges the node.
- **Negative voltage** ($-0.7 \leq V_m < 0$): When $V_m < 0$ occurs during node discharge at C_m , forward biasing of the drain-substrate junction may occur; the ratio $R_{leak} = V/I$ may appear with a negative sign according to the direction of the current. This does not imply a physically negative resistance; it is an undesirable transient condition in the circuit.

This transient negative-voltage behavior at the C_m node arises from charge redistribution effects, primarily driven by the gate-to-drain parasitic capacitance of M1 during the fast transition of the V_{gs1} pulse. When the reset signal switches abruptly, capacitive coupling between the gate and drain momentarily injects charge into the membrane node, pulling the V_m voltage below ground. This effect is exacerbated when the effective membrane capacitance is dominated by small parasitic components. By increasing C_m through the addition of a small explicit MiM capacitor, the relative impact of this parasitic charge injection is significantly reduced, and the transient negative excursion at the membrane node effectively vanishes. In this case, the larger total capacitance absorbs the injected charge, preventing appreciable voltage undershoot. When a transient negative voltage appears at the drain of M1, the drain-substrate p-n junction may become forward-biased. Since this junction begins to conduct at approximately 0.7 V, corresponding to the typical forward voltage of a silicon diode, the maximum negative excursion of the drain voltage is inherently limited to approximately $-V_f \approx -0.7$ V. Forward conduction clamps the membrane potential and prevents further decrease.

The leakage resistance at the drain of M1, denoted as R_{leak} , varies as a function of the membrane voltage. When the membrane voltage V_m approaches zero, the effective resistance is relatively low. As V_m increases during the integration phase, R_{leak} correspondingly increases. When the membrane potential reaches the firing threshold, the operating regime of M1 changes abruptly: the device exits the subthreshold region and rapidly transitions into strong inversion. This regime change enables a much larger drain-source current, which efficiently discharges the charge accumulated at the V_m node, thereby resetting the membrane

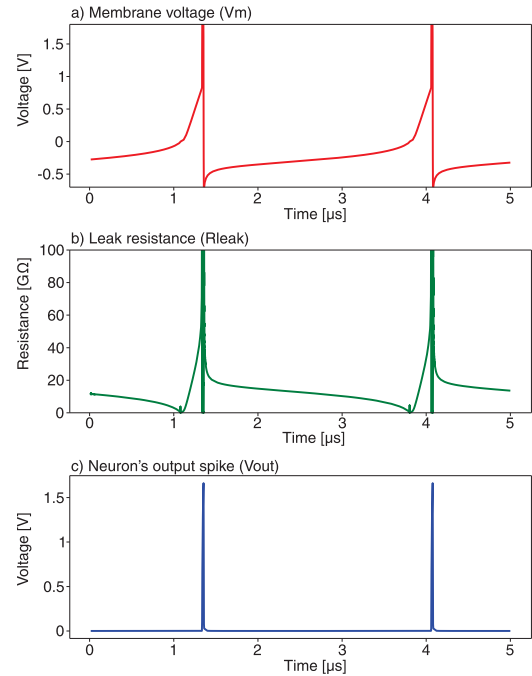


FIGURE 7. Transient waveforms of the LIF neuron: (a) membrane voltage V_m , (b) effective leakage resistance R_{leak} , and (c) output spike waveform; illustrating subthreshold integration followed by saturation-driven reset.

voltage toward zero and initiating the next integration cycle.

In the R_{leak} waveform (Fig. 7(b)), two distinct regimes can be identified. Initially, R_{leak} exhibits a gradual evolution with relatively low resistance values, consistent with the slow and controlled charging of the membrane capacitor observed over the same time interval in V_m . This behavior corresponds to subthreshold operation, in which M1 provides a weak but nonzero conduction path to ground during the integration phase.

As the membrane voltage increases, R_{leak} rises exponentially, reflecting the strong dependence of the subthreshold drain current on the drain–source voltage. It is important to recall that, between spike events, the gate voltage of M1 remains essentially at 0 V. Although the transistor can be regarded as “off” from a digital perspective, it never fully turns off; therefore, the subthreshold conduction model remains valid throughout the integration interval.

Over the practical operating range, from V_{reset} up to the inverter switching threshold $V_{th(lif)}$, the variation of R_{leak} can be reasonably approximated as quasi-linear. This observation justifies the linearization of $R_{leak}(V_m)$ adopted in the subsequent analysis, enabling an analytical estimate of the average leak resistance used in the firing-frequency derivation. The upper bound of the membrane voltage is ultimately set by the inverter switching threshold, which defines the neuron firing level and marks the transition to the second stage of circuit operation.

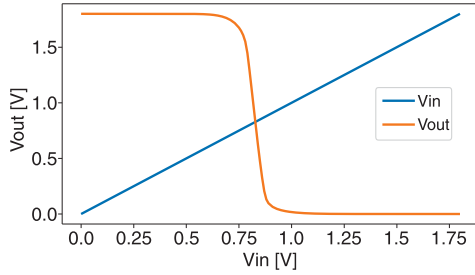


FIGURE 8. Voltage-transfer characteristic (VTC) of the inverter formed by transistors M2–M3, showing the switching voltage V_c , which defines the effective LIF firing threshold $V_{th(lif)}$.

C. INVERTER SWITCHING THRESHOLD $V_{th(lif)}$

In this stage, the LIF neuron threshold voltage, also referred to as the inverter switching voltage, is derived from the MOS transistor conduction-current expressions. The inverter consists of a complementary pair of MOS transistors, one p-type and one n-type. The switching voltage corresponds to the operating point at which both transistors conduct simultaneously [17]. On the voltage-transfer characteristic (VTC), this condition is identified at the intersection of the input-output curves (Fig. 8), where the drain currents of the p- and n-type devices are equal, i.e., $I_{dp} = I_{dn}$ as expressed in (12). At this operating point, both transistors reside in the saturation region, which defines the inverter's transition behavior and establishes the effective firing threshold of the LIF neuron.

$$I_{dp} = I_{dn}, \quad (10)$$

$$k_0 = \mu_0 C_{ox} \frac{W}{L}, \quad (11)$$

$$\frac{1}{2} k_p (V_{sg} - |V_{th}|)^2 = \frac{1}{2} k_n (V_{gs} - V_{th})^2, \quad (12)$$

at the inverter switching point, it is necessary that $V_{in} = V_{out}$. We denote this common value as V_c (the switching voltage), which defines the neuron's effective threshold $V_{th(lif)}$, and it should not be confused with the device threshold V_{th} . From this, we obtain:

$$k_p (V_{dd} - V_c - |V_{th}|)^2 = k_n (V_c - 0 - V_{th})^2, \quad (13)$$

$$\sqrt{k_p} (V_{dd} - V_c - |V_{th}|) = \sqrt{k_n} (V_c - V_{th}), \quad (14)$$

$$\sqrt{k_p} V_{dd} - \sqrt{k_p} V_c - \sqrt{k_p} |V_{th}| = \sqrt{k_n} V_c - \sqrt{k_n} V_{th}, \quad (15)$$

$$V_c = \frac{\sqrt{k_p} (V_{dd} - |V_{th}|) + \sqrt{k_n} V_{th}}{\sqrt{k_n} + \sqrt{k_p}} = V_{th(lif)}. \quad (16)$$

Equation (16) provides the inverter switching voltage, which, in our implementation, is the effective LIF threshold. This implies that the threshold can be tuned to tailor the neuron's behavior. At first glance, one might expect a lower threshold to accelerate firing, whereas a higher threshold would delay it.

However, because transistor M1 operates in the subthreshold region, its behavior is highly sensitive to the parasitic

capacitances present at the membrane node. As a result, any modification in the sizing of transistors M2 and M3 (i.e., their W/L ratios) alters not only the inverter switching threshold but also the effective capacitive loading seen at the membrane node, directly affecting the integration time constant. Consequently, increasing the inverter switching threshold $V_{th(lif)}$ alone does not necessarily lead to a longer integration interval or increased firing latency. Instead, the neuron dynamics are primarily governed by the effective membrane time constant $\tau = R_{leak} C_m$, which depends on both the subthreshold leakage resistance of M1 and the total capacitance at the membrane node.

In practice, $V_{th(lif)}$ must therefore be co-designed with the membrane capacitance, comprising parasitic and optional explicit components, and with the subthreshold leakage characteristics that define R_{leak} . This co-design approach enables controlled tuning of the firing frequency through time-constant engineering rather than relying solely on threshold adjustment.

D. PARASITIC CAPACITANCE CONSIDERATIONS

To complete the analysis, it is necessary to determine the value of C_m , which represents the equivalent capacitance at the integration (membrane) node. This capacitance is formed by the cumulative parasitic contributions of all transistors connected to the node, as well as by any external elements responsible for delivering the excitation current. It is important to emphasize that, when operating in the subthreshold region, transistor M1 becomes highly sensitive to capacitive effects at the membrane node. While subthreshold operation offers substantial benefits, most notably, a significant reduction in energy consumption, it also increases susceptibility to process variations, parasitic loading, and noise coupling. As a result, an accurate estimation of C_m is essential for predicting the neuron's temporal dynamics and firing behavior.

Considering only transistors M1, M2, and M3, the total membrane capacitance can be approximated as:

$$C_t = C_{gdM1} + C_{bdM1} + C_{gsM2} + C_{gdM2} + C_{gbM2} + C_{gsM3} + C_{gdM3} + C_{gbM3} + C_m, \quad (17)$$

where C_t denotes any explicit capacitor connected at the node. It follows that larger transistor sizes lead to larger parasitic capacitances; therefore, a balance between device sizing and parasitics must be established to achieve the desired circuit behavior.

A practical issue arises when an ideal current source is connected directly to the integration node, as illustrated in Fig. 9, since such a source does not contribute any parasitic capacitance. Although this idealization may be acceptable for a first-order analysis, the excitation currents of interest lie in the nanoampere range. Under these conditions, even small variations in the effective capacitance at the membrane node can lead to noticeable changes in the charging dynamics and, consequently, in the neuron firing frequency, as illustrated in Fig. 10.

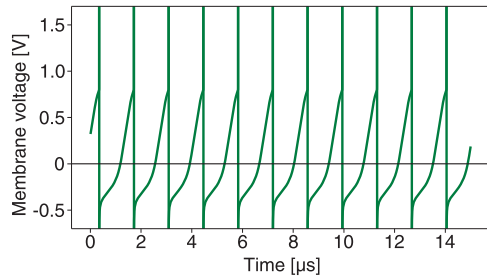


FIGURE 9. Response of the LIF circuit with an ideal current source, $I_{ex} = 23.8$ nA.

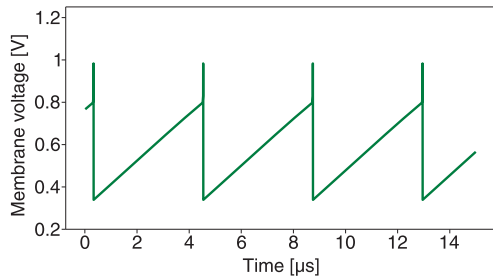


FIGURE 10. Membrane-voltage response when a capacitive load is present, showing modified integration dynamics and firing period due to additional capacitive loading. With $I_{ex} = 23.8$ nA and $C_m = 0.2$ pF.

At this stage, the design can follow one of two complementary strategies:

- 1) Introduce a small additional capacitance as an explicit design parameter to stabilize the integration node.
- 2) Isolate the integration node using a dedicated input element to decouple it from external parasitic effects.

In this work, we deliberately combine both strategies because they address different (and complementary) sources of variability at the integration node. Adding a small explicit C_m provides a well-defined capacitive reference that reduces the relative impact of parasitic charge injection (e.g., via gate-drain coupling during fast switching) and stabilizes the membrane trajectory, preventing undesired voltage undershoot. In parallel, introducing a dedicated input (isolation) element decouples the membrane node from the excitation source and its associated parasitics, ensuring that variations in the current-delivery circuitry do not unintentionally modulate the effective node capacitance.

Therefore, by combining isolation and explicit capacitance, we turn the membrane dynamics into a controlled, designable time constant $\tau = R_{leak}C_m$, improving the predictability of the firing frequency, robustness against parasitic fluctuations, and compatibility with integration in larger mixed-signal systems.

Our proposal is to incorporate an additional transistor that acts as a current-source buffer, delivering the required excitation current in a controlled manner, and to add a

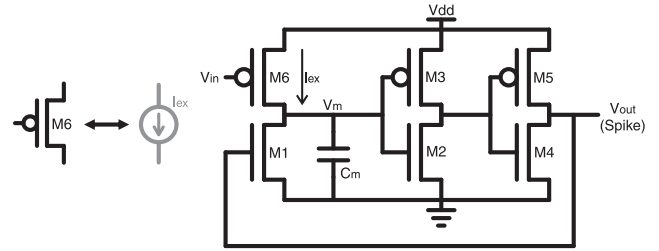


FIGURE 11. Modified LIF circuit including an isolation transistor (M6) and a small explicit membrane capacitor C_m , used to decouple the integration node from external parasitics and mitigate charge-injection effects.

compensation capacitor (C_m) to prevent undesired current flows, as shown in Fig. 11. This keeps the membrane voltage within the supply range (from 0 to V_{dd}). It is worth noting that the proposed circuit shares the same general structure as other implementations reported in the literature [18], [19], differing only in the placement of the capacitor and the supply voltage levels. This allows tighter control of the membrane voltage, which in turn enables a more accurate estimate of the firing frequency, as shown in (21).

Although a transistor is proposed as the isolation element, any component that provides capacitive decoupling could be used in its place. The isolation transistor also adds its own capacitances to the node, leading to the modified expression in (18):

$$C_t = C_{gdM1} + C_{bdM1} + C_{gsM2} + C_{gdM2} + C_{gbM2} + C_{gsM3} + C_{gdM3} + C_{gbM3} + C_{gsM6} + C_{bsM6} + C_m, \quad (18)$$

where transistor $M6$ denotes the isolation device and C_m denotes the explicit membrane capacitor (if present).

Although the isolation transistor is not strictly required for circuit functionality, its inclusion significantly enhances robustness by mitigating the influence of parasitic capacitances at the integration node. The device effectively acts as a buffer, decoupling the membrane node from the capacitances associated with the excitation source. As a result, the charge and discharge dynamics of the membrane capacitor become more predictable, enabling tighter control of the effective time constant and, consequently, of the neuron firing behavior. In addition, the explicit capacitor allows for tuning the neuron's firing frequency. This capacitor is connected between the integration node and the reference potential (GND), and its value can be adjusted to modify the circuit time constant.

With these modifications, the response of the LIF circuit, including both the isolation transistor and the compensation capacitor, is shown in Fig. 12. The resulting waveforms exhibit more stable and well-controlled membrane dynamics, with the membrane voltage remaining within the supply range. This improved behavior facilitates reliable integration with other circuit blocks and significantly enhances robustness against variations in parasitic capacitance.

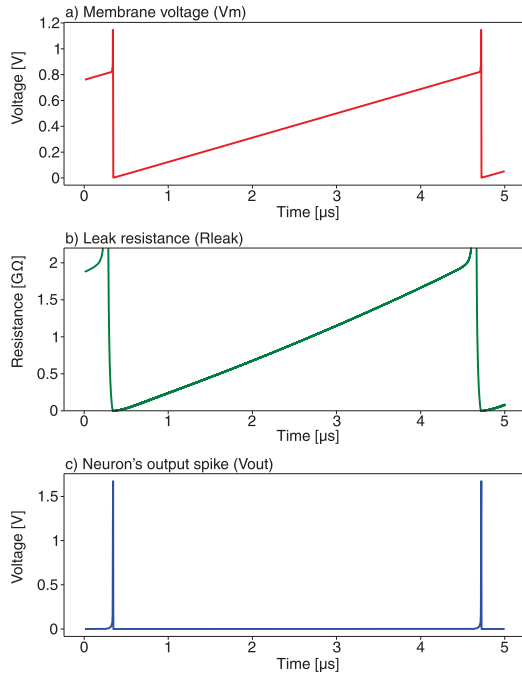


FIGURE 12. Transient response of the modified LIF circuit with isolation and compensation capacitance: (a) membrane voltage, (b) effective leakage resistance, and (c) output spike waveform, demonstrating improved stability and controlled dynamics, with $I_{ex} = 23.8nA$.

E. FIRING FREQUENCY

The firing frequency refers to the number of pulses generated per unit time and is directly related to the circuit time constant, which depends on the resistance and capacitance at the integration node. To determine the time interval required for this model to reach the threshold voltage (19), we impose the condition $v_m = v_{th(lif)}$:

$$v_{th(lif)} = R_{leak} I_{ex} + \frac{v_{reset} - R_{leak} I_{ex}}{e^{t/(R_{leak} C_m)}}. \quad (19)$$

Solving for t yields the time required for the membrane voltage to reach the firing threshold $v_{th(lif)}$, as given in (20):

$$t = R_{leak} C_m \ln \left(\frac{v_{reset} - R_{leak} I_{ex}}{v_{th(lif)} - R_{leak} I_{ex}} \right). \quad (20)$$

Recalling that the firing frequency is defined as the inverse of the oscillation period, the corresponding firing-rate expression is obtained as (21):

$$f = \frac{1}{R_{leak} C_m \ln \left(\frac{v_{reset} - R_{leak} I_{ex}}{v_{th(lif)} - R_{leak} I_{ex}} \right)}. \quad (21)$$

From (21), it is evident that the firing frequency depends primarily on the excitation current I_{ex} and the effective leakage resistance R_{leak} , while the remaining parameters are treated as design-time constants.

Because R_{leak} varies over the integration phase as a function of the membrane voltage, a practical approximation consists of computing its average value over the relevant

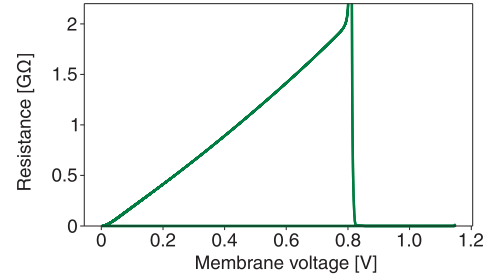


FIGURE 13. Effective leakage resistance R_{leak} versus membrane voltage V_m over the operating range from V_{reset} to $V_{th(lif)}$, illustrating the quasi-linear region used for analytical approximation, with $I_{ex} = 23.8nA$.

membrane-voltage range, as described in (22):

$$\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx, \quad (22)$$

It is important to note that R_{leak} is defined as a function of the membrane voltage V_m , while V_m itself depends on R_{leak} through the integration dynamics. This mutual dependence introduces an implicit coupling between the two quantities.

Because this interdependence precludes a straightforward closed-form solution, we approximate the dependence of R_{leak} on V_m with a linear relationship (Fig. 13) over the operating range of interest.

This simplification preserves the essential model behavior while enabling an analytical treatment (23) (remember that $V_{reset} \geq 0$).

$$R_{leak}(V_m) = m V_m + b, \quad V_{reset} \leq V_m \leq V_{th(lif)}, \quad (23)$$

where,

$$m = \frac{R_{leak}(V_{m2}) - R_{leak}(V_{m1})}{V_{m2} - V_{m1}}, \quad (24)$$

$$b = R_{leak}(V_m) - m V_m. \quad (25)$$

Under this linearization, the average leak resistance over the integration swing is:

$$\bar{R}_{leak} = \frac{1}{V_{th(lif)} - V_{reset}} \int_{V_{reset}}^{V_{th(lif)}} (m V_m + b) dV_m, \quad (26)$$

$$\bar{R}_{reset} = \left(\frac{1}{V_{th(lif)} - V_{reset}} \right) \left(\frac{m V_m^2}{2} + b V_m \right) \Big|_{V_{reset}}^{V_{th(lif)}}, \quad (27)$$

$$\bar{R}_{leak} = \left(\frac{1}{V_{th(lif)} - V_{reset}} \right) \left(\frac{m}{2} (V_{th(lif)}^2 - V_{reset}^2) + b (V_{th(lif)} - V_{reset}) \right), \quad (28)$$

$$\bar{R}_{leak} = \frac{m}{2} (V_{th(lif)} + V_{reset}) + b. \quad (29)$$

If $V_{reset} = 0$, (29) simplifies to (30):

$$\bar{R}_{leak} = \frac{m V_{th(lif)}}{2} + b, \quad (30)$$

finally, the firing frequency can be written as (31):

$$f = \frac{1}{\bar{R}_{leak} C_m \ln \left(\frac{v_{reset} - \bar{R}_{leak} I_{ex}}{v_{th(lif)} - \bar{R}_{leak} I_{ex}} \right)}, \quad (31)$$

TABLE 1. Average leakage resistance $\bar{R}_{\text{leak}}$, excitation current I_{ex} , and their product for different operating points, confirming that $\bar{R}_{\text{leak}} \cdot I_{\text{ex}}$ remains approximately constant and above $V_{\text{th(lif)}}$.

m	b	R (M Ω)	I_{ex} (nA)	$R \cdot I$
5.19×10^8	-1.40×10^7	193.9	118.0	22.88
6.56×10^8	-1.85×10^7	244.0	93.10	22.72
8.51×10^8	-2.17×10^7	318.7	71.40	22.76
1.15×10^9	-3.06×10^7	431.5	52.60	22.69
1.65×10^9	-4.44×10^7	616.6	36.70	22.63
2.54×10^9	-6.58×10^7	952.7	23.80	22.68
4.40×10^9	-1.18×10^8	1644.5	13.80	22.69
9.16×10^9	-2.45×10^8	3422.1	6.61	22.62
2.59×10^{10}	-2.59×10^8	9709.7	2.32	22.53

and, under the same approximation $V_{\text{reset}} = 0$, it reduces to (32):

$$f = \frac{1}{\bar{R}_{\text{leak}} C_m \ln \left(\frac{-\bar{R}_{\text{leak}} I_{\text{ex}}}{V_{\text{th(LIF)}} - \bar{R}_{\text{leak}} I_{\text{ex}}} \right)}. \quad (32)$$

Once the firing-frequency expression is available, it can be used to analyze the LIF circuit's behavior as a function of its design parameters. This expression enables systematic tuning of the excitation current I_{ex} , the membrane capacitance C_m , and the threshold voltage $V_{\text{th(LIF)}}$, together with the reset level V_{reset} when nonzero, as well as the effective average leakage resistance $\bar{R}_{\text{leak}}$, in order to achieve a desired firing rate. Such tunability is essential for applications that require precise control of neuronal dynamics.

The frequency expression in (32) is valid when the reset voltage is nonnegative ($V_{\text{reset}} \geq 0$), a condition under which R_{leak} exhibits an approximately linear dependence over the integration range. Moreover, when the devices connected to the integration node are aggressively scaled down, the explicit membrane capacitor may dominate the charge-discharge dynamics. In this case, the firing frequency can be accurately estimated using a simplified capacitor-dominated model.

Another important aspect of (32) is the constraint imposed by the natural logarithm, which requires its argument to be strictly positive. Consequently, it must be ensured that the product $R_{\text{leak}} \cdot I_{\text{ex}}$ always exceeds the firing threshold $V_{\text{th(lif)}}$. To verify this condition, the average leakage resistance $\bar{R}_{\text{leak}}$ was computed for each excitation current using (30). The results show that the product $\bar{R}_{\text{leak}} \cdot I_{\text{ex}}$ remains approximately constant and consistently exceeds $V_{\text{th(LIF)}}$, as summarized in Table 1. This confirms that the logarithm's argument is always positive over the considered operating range.

Another way to compute the average resistance analytically is to use (9). Owing to the linear behavior of the resistance, the average of a linear function over an interval equals its value at the midpoint. Thus, it suffices to substitute $V_m = (V_{\text{th(LIF)}} + V_{\text{reset}})/2$ into (9) to obtain (33):

$$\bar{R}_{\text{leak}} \approx \frac{V_{\text{th(lif)}} + V_{\text{reset}}}{2I_0 \exp\left(-\frac{V_{\text{th}}}{nV_t}\right) \left(1 - \exp\left(-\frac{V_{\text{th(lif)}} + V_{\text{reset}}}{2V_t}\right)\right)}, \quad (33)$$

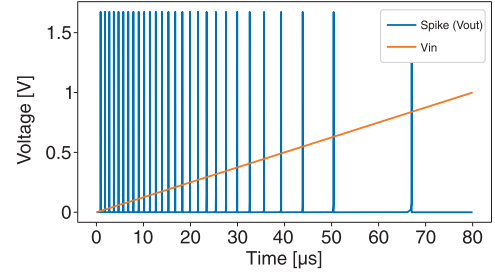


FIGURE 14. LIF neuron response to a ramp input voltage, from 0 to 1 V, applied to the excitation device, demonstrating spike generation over a wide dynamic range of excitation currents.

finally, if $V_{\text{reset}} = 0$, the equation reduces to (34).

$$\bar{R}_{\text{leak}} \approx \frac{V_{\text{th(lif)}}}{2I_0 \exp\left(-\frac{V_{\text{th}}}{nV_t}\right) \left(1 - \exp\left(-\frac{V_{\text{th(lif)}}}{2V_t}\right)\right)}. \quad (34)$$

By obtaining the average resistance from Eq. (34), a resistive value of 964M Ω is obtained. Referring to Table 1, we note that the average resistance lies within the measurement range, very close to the resistive value for $i_{\text{ex}} = 23.8$ nA. This is also visible in Fig. 13, where the mean resistance is approximately 1 G Ω . As mentioned before, since the product $(R \cdot I)$ varies very little, it can be used with Eq. (31) to calculate the firing frequencies that this circuit can deliver.

IV. RESULTS AND DISCUSSION

This section presents the obtained results and their comparison with manufacturable-technology simulations, emphasizing the agreement with theoretical predictions. It also analyzes the energy-latency trade-offs, silicon area, and comparison with previous implementations.

A. RESULTS

To evaluate the performance of the proposed dual-mode CMOS (complementary metal-oxide-semiconductor) LIF neuron, we ran circuit-level simulations using Xschem linked with the SkyWater SKY130 process design kit (PDK). Transient analyzes were performed with a SPICE-class simulator with a supply voltage of $V_{\text{dd}} = 1.8$ V (Fig. 11).

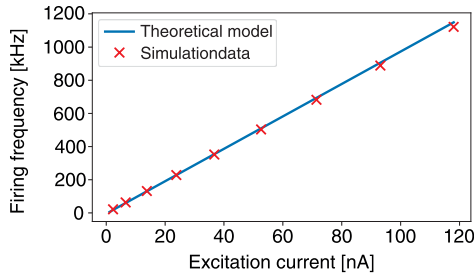
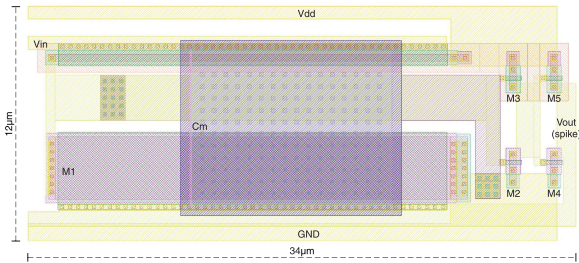
Figure 14 presents the membrane response for various gate bias values applied to the input device, where a ramp-like stimulus drives the LIF circuit. This circuit generates spikes in the kilohertz to megahertz range, which are suitable for activating resistive memories.

Table 2 summarizes the results for different gate voltages (applied to M6), reporting the corresponding excitation current and firing frequency.

A comparative analysis between simulated frequencies and the theoretical estimate given by (32) is shown in Fig. 15. The plot evidences good agreement, supporting the validity of the proposed model and its predictive capability for the LIF circuit.

TABLE 2. Measured excitation current flowing through M6, firing frequency, and average power consumption for different gate-bias values applied to the gate of the isolation transistor (M6).

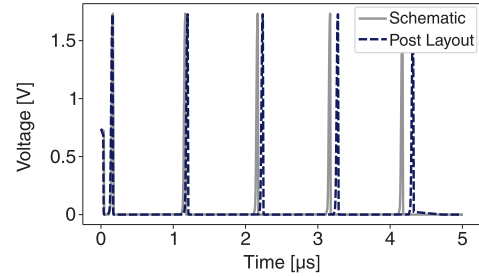
V_{M6} (V)	I_{ex} (nA)	f (kHz)	P (nW)
0.0	118.0	1121.65	44.33
0.1	93.10	888.49	35.11
0.2	71.40	682.27	26.96
0.3	52.60	503.42	19.90
0.4	36.70	352.15	13.92
0.5	23.80	228.43	9.03
0.6	13.80	132.18	5.22
0.7	6.61	63.42	2.51
0.8	2.32	22.15	0.88

**FIGURE 15.** Comparison between analytically predicted firing frequency and SPICE simulation results, showing good agreement across the explored excitation-current range.**FIGURE 16.** Layout view of the proposed dual-mode CMOS LIF neuron implemented in the SKY130 process. The compact layout confirms the low-area footprint of the design, which is $408\mu\text{m}^2$.

The circuit layout is shown in Fig. 16. The mask design was implemented in KLayout following the foundry design rules (DRC). The total active area is $408\mu\text{m}^2$.

The explicit (or effective) membrane capacitor is relatively small compared with the transistor area because the required capacitance is on the order of a few femtofarads; thus, no large capacitor is needed. By contrast, transistors M1 and M6 dominate the silicon area: M1 is sized to set a large effective leak resistance, and M6 is sized to reduce the excitation current via isolation/buffering. With further design-space exploration (e.g., sizing heuristics or simple optimization), the total area may be reduced; increasing the explicit C_m allows for smaller devices while preserving the target time constant.

This physical design is suitable for various integration applications, as compact LIF neurons can be arranged in a matrix with minimal area overhead. In neuron circuits, capacitors often limit the number of neurons that can be

**FIGURE 17.** Comparison between schematic-level and post-layout transient simulations, highlighting the modest frequency reduction caused by additional parasitic capacitances after layout extraction.

placed on a chip. By utilizing parasitic capacitances and maintaining a small explicit membrane capacitance (C_m), it is possible to increase neuron density.

Finally, Fig. 17 contrasts the schematic-level and post-layout simulations. The post-layout model exhibits a slightly lower firing frequency, attributable to increased parasitic capacitances that lengthen the membrane charge time and reduce the oscillation frequency. In our case, post-layout simulation achieves 961 kHz versus 998 kHz in the schematic view simulation. Despite this difference, the similarity in oscillation frequency is 96.33%, indicating that the layout preserves the intended functionality and that the residual deviation stems from non-critical parasitic effects.

An additional performance aspect is energy consumption. The energy per spike event can be approximated by:

$$E = \frac{1}{2} C_m V_{th(LIF)}^2, \quad (35)$$

where C_m is the membrane capacitance ($C_m = 123.5$ fF in our circuit) and $V_{th(LIF)}$ is the peak membrane voltage during charging ($V_{th(LIF)} \approx 0.8$ V). Using these values, we obtain $E \approx 35.9$ fJ. The average power is then:

$$P = E \cdot f, \quad (36)$$

so, Table 2 also reflects a power–frequency relation consistent with subthreshold-optimized operation: ultra-low currents and reduced energy per spike.

Lastly, Table 3 compares the proposed neuron with recent implementations. While several designs report lower or higher absolute energy depending on operating point and load, our approach combines a low device count, no external amplification for I/O compatibility (full swing), and predictable tuning via C_m and $V_{th(lif)}$.

Finally, Table 4 summarizes the values employed in the development of this study.

B. DISCUSSION

1) ENERGY-LATENCY AND ROBUSTNESS

The results demonstrate that the proposed dual-mode topology leverages subthreshold integration to reduce the energy consumed per spike in the low-frequency regime while

TABLE 3. Comparison of the proposed LIF neuron with recent neuromorphic implementations, highlighting operating frequency, energy per spike, transistor count, operation region, and the need for output amplification.

Reference	Neuron type	Frequency range	Energy per spike	Transistor count	Operation region	Requires I/O amplification
[3]	ML Biomimetic	1.2 kHz	78.3 fJ	8	Subthreshold	Yes
[3]	ML Simplified	25 kHz	4 fJ	6	Subthreshold	Yes
[5]	LIF	kHz	4.3 pJ	14	Saturation	No
[6]	LIF	kHz	200 pJ	13+	Saturation	No
[7]	LIF (Analog)	1 MHz	2 pJ	24+	Saturation	No
[7]	LIF (Digital)	1 MHz	41.3 pJ	–	Saturation	No
[9]	LIF	52.6 MHz	0.67 fJ	8	Subthreshold	Yes
[18]	AH	15.7 kHz	2 fJ	5	Subthreshold	Yes
This work	LIF	1.1 MHz	35.9 fJ	5–6	Mixed	No

TABLE 4. Summary of device dimensions, electrical parameters, and model constants used in the analytical derivation and circuit simulations.

Parameter	Value
M1	$3\mu\text{m}/10\mu\text{m}$
M2	$450\text{nm}/150\text{nm}$
M3	$450\text{nm}/150\text{nm}$
M4	$450\text{nm}/150\text{nm}$
M5	$450\text{nm}/150\text{nm}$
M6	$450\text{nm}/99\mu\text{m}$
C_m	123.5fF
$V_{\text{th(lif)}}$	0.8285V
V_{reset}	0V
$\bar{R}$	949674625Ω
V_t	25.86mV
$\mu_n C_{\text{ox}}$	$257.24\mu\text{A}/\text{V}^2$
n	3.6
V_{th}	0.537756V

preserving a full output voltage swing through its saturation-region stage. This combination eliminates the need for external amplifiers or level shifters and enables seamless *plug-and-play* integration with saturation-range analog and digital blocks.

The associated trade-off is a modest increase in power consumption relative to strictly subthreshold designs. In exchange, the neuron achieves significantly improved I/O robustness while operating from a single supply, the same V_{DD} rail as the host circuit, and maintains a very low transistor count (5T in the baseline configuration, 6T with isolation). These characteristics make the proposed design particularly well-suited for dense neuron arrays and mixed-regime neuromorphic systems.

2) MODEL-BASED FREQUENCY TUNING (DESIGN-BY-SPECIFICATION)

Unlike much of the existing literature, which often reports operating frequencies without providing a clear methodology for tuning them, this work derives an analytical framework that explicitly links the inverter trip point $V_{\text{th(LIF)}}$, the effective membrane capacitance C_m , and the averaged leakage resistance $\bar{R}_{\text{leak}}$ to the firing rate through (32), with $\bar{R}_{\text{leak}}$ obtained from (29) and (30).

This set of equations enables a *design-by-specification* approach: given a target firing frequency, the designer can select C_m , size transistors M2–M3 to set $V_{\text{th(lif)}}$, and choose the excitation current I_{ex} to meet performance objectives under area and energy constraints. In contrast to prior implementations summarized in Table 3, which primarily report achievable frequency and energy figures without an explicit tuning methodology, the proposed framework provides a systematic, predictive path from specifications to circuit parameters, facilitating fair comparisons and practical reuse across different operating points.

V. CONCLUSION AND FUTURE WORK

Finally, in this section, the main findings are synthesized and potential improvements aimed at specific applications are presented.

A. CONCLUSIONS

This work presents the design and analysis of a compact, dual-mode CMOS LIF neuron that leverages a single subthreshold-operated device for ultra-low-power integration, combined with saturation-region transistors that provide robust thresholding, reset, and full-swing input/output operation. The proposed architecture achieves low energy per spike while preserving a practical output voltage swing, enabling direct interfacing with saturation-range analog and digital blocks without the need for external amplification stages or multiple supply rails.

Compared with neuron circuits implemented purely in saturation, the proposed circuit reduces energy consumption for low-frequency workloads by exploiting subthreshold conduction and the membrane node's equivalent parasitic capacitance, all with a *very low transistor count* (5T baseline, 6T with isolation).

Relative to state-of-the-art subthreshold-only LIF designs, our neuron maintains a similarly small device budget while slightly increasing consumption to gain two pragmatic advantages: (i) elimination of dedicated output amplifiers or level shifters, and (ii) single-supply operation, which substantially simplifies system-level integration. These characteristics make the proposed neuron both novel and

practical for dense neuromorphic arrays and mixed-regime systems.

From an analytical standpoint, we derived a closed-form expression for the firing frequency that explicitly links the inverter trip point $V_{th(LIF)}$, the effective membrane capacitance C_m , and the averaged leakage resistance $\bar{R}_{leak}$ to the observed spiking rate. Circuit-level simulations using the SKY130 PDK validate the model across the intended operating region, showing good agreement between predicted and simulated firing frequencies and consistent energy-latency trends. Post-layout results further confirm correct functionality, with the expected frequency reduction due to additional parasitic capacitances remaining within acceptable limits.

B. FUTURE WORK

Several directions can further improve robustness and tunability:

- **Digitally controlled capacitor bank:** add an on-chip C_m bank with simple digital control to trim the firing frequency post-fabrication.
- **Control and extensibility:** because the neuron's input and output stages operate in saturation, it is straightforward to incorporate control features and mixed-signal hooks. For example, an inhibitory gate (a transistor that momentarily blocks current into the integrate node) can be added to suppress spiking; multiple input transistors can be placed in parallel to accept several excitation signals; and, more generally, the saturation-mode interface simplifies the integration of additional digital or analog circuitry as needed.

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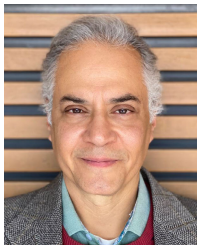
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